

Extended Formulations

Central Question: Can a polytope $P \subseteq \mathbb{R}^n$

with $\exp(n)$ facets, be described
as the projection of $P_E \subseteq \mathbb{R}^N$,

$N = \text{poly}(n)$, and P_E has $\text{poly}(n)$ facets?

Significance: If $P = AP_E$ for A an $n \times N$ matrix

$$\begin{array}{ll} \text{then;} & \min: c^T x \\ & \text{st: } x \in P \end{array} = \begin{array}{ll} \min: & (A^T c)^T X \\ \text{st:} & X \in P_E \end{array}$$

Example: The Permutahedron

S^n denote the $n!$ permutations on $\{1, \dots, n\}$.

Let's define: $P_{\text{perm}}^{(n)} \triangleq \text{conv} \{ (\pi(1), \pi(2), \dots, \pi(n)) : \pi \in S^n \}$

Ex: $\underline{P_{\text{perm}}^{(3)}} = \text{conv} \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \right\}$

Claim: $P_{\text{perm}}^{(n)} = \left\{ x \geq 0, \sum x_i = \binom{n+1}{2}, \sum_{i \in S} x_i \leq \binom{|S|+1}{2}, S \subseteq [n] \right\}$

/ exp'ly many facets.

$= \left\{ x \geq 0, x([n]) = \binom{n+1}{2}, x(T) \leq \binom{n+1}{2} - \binom{n-|T|+1}{2}, \forall T \subseteq [n] \right\}$

Note: $P_{\text{perm}}^{(n)} = \left\{ x \geq 0, x([n]) = \binom{n+1}{2}, \underline{x(T) \leq f(T)}, \forall T \subseteq [n] \right\}$

where $f(T)$ is submodular.

This is therefore a special case of: Polymatroid

The Permutohedron - continued -

For any $P = \{x: x(s) \leq f(s), \forall s \in E\}$

for f is submodular, $\forall c \geq 0$, the

problem:

$\max: c^T x$
 $\text{s.t.: } x \in P$ has a solution x^* given as follows:

Re-order coordinates $1, \dots, |E|$ s.t. $c_1 \geq c_2 \geq \dots \geq c_n \geq 0$

$$\text{Then: } x^* = \begin{pmatrix} f(e_1) \\ f(e_1, e_2) - f(e_1) \\ \vdots \\ f(e_1, \dots, e_{|E|}) - f(e_1, \dots, e_{|E|-1}) \end{pmatrix}$$

Note: If f is integer-valued submodular function, then P has integral vertices. Thus: P_{perm} is integral. Hence our facet desc. is correct.

Note: $\hat{f}(c) = \max_{x \in P} c^T x$, $\hat{f}$ is the Lovasz ext. of f for $c \in [0, 1]^E$. // EX.

Permutahedron Extended Formulation

Now we give an extended formulation with $n^2 + 3n$ constraints in $n^2 + n$ dimensions

Given $\pi \in S^n$, Y_π $n \times n$ matrix

$$Y_\pi(i,j) = \begin{cases} 1 & \text{if } j = \pi(i) \\ 0 & \text{o.w.} \end{cases}$$

$$x_\pi(i) = \sum_j j \cdot Y_\pi(i,j)$$

$$\text{Conv} \{ Y_\pi : \pi \in S^n \} = \left\{ Y \geq 0, \sum_j Y_{ij} = 1, \sum_i Y_{ij} = 1 \quad \forall i,j \right\}$$

= Bipartite matching polytope on $K_{n,n}$



$$P_{\text{perm}}^{(n)} = \left\{ x \in \mathbb{R}^n : \exists Y \geq 0, \sum_j Y_{ij} = \sum_i Y_{ij} = 1, \forall i,j, x_i = \sum_j j Y_{ij} \quad \forall i \right\}$$

$N = n^2 + n$ dim, with $n^2 + 3n$ constraints